

AI/ML Task 3 — Model Validation, Overfitting Control & Hyperparameter Tuning

House Price Prediction — California Housing Dataset

1. Objective

Task 3 extends the Task-2 pipeline from a single train-test evaluation to a full professional ML workflow: detecting overfitting, applying 5-fold cross-validation for reliable performance estimation, and tuning hyperparameters systematically with GridSearchCV. The goal is **model reliability**, not just accuracy.

2. Dataset & Methodology

Dataset: California Housing (20,433 records, 8 features, same as Task-2). Pipeline: StandardScaler → 80/20 train-test split (random_state=42) → overfitting detection → 5-fold cross-validation → GridSearchCV tuning on Decision Tree and Ridge Regression → final comparison.

Key concepts applied:

- **Overfitting detection:** Train vs test RMSE gap on unconstrained Decision Tree.
- **Cross-validation:** 5-fold CV with neg_root_mean_squared_error scoring gives stable, split-independent estimates.
- **GridSearchCV:** Exhaustive search over $\text{max_depth} \in \{3, 5, 7, 10\} \times \text{min_samples_split} \in \{2, 5, 10\}$ for DT; $\alpha \in \{0.01, 0.1, 1, 10, 100\}$ for Ridge.
- **Bias-Variance analysis:** Train/test RMSE tracked across all depth values to visualise the underfitting → sweet-spot → overfitting curve.

3. Overfitting Analysis — Unconstrained Decision Tree

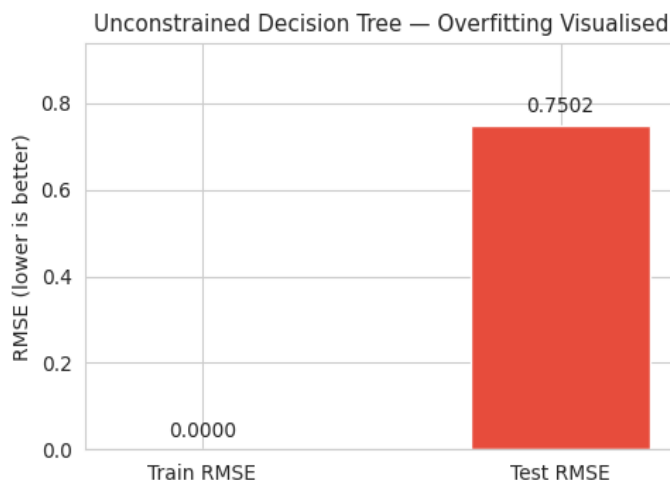


Figure 1: Train vs Test RMSE for unconstrained Decision Tree.

An unconstrained Decision Tree (no max_depth) perfectly memorises training data: **Train RMSE = 0.0000, Test RMSE = 0.7502 — a gap of 0.7502**. This is severe overfitting. The model learned noise in the training set rather than generalizable patterns, making it useless for production.

4. Cross-Validation — Baseline Model Comparison

Model	CV RMSE (5-fold mean)	CV RMSE Std
Linear Regression (baseline)	0.7460	0.0060
Ridge Regression ($\alpha=1$)	0.7460	0.0060

Decision Tree (unconstrained)	0.8678	0.0200
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Cross-validation reveals the unconstrained DT CV RMSE (0.8678) is far worse than its test RMSE (0.7502), showing the single test split was optimistic. Linear and Ridge models are stable (CV RMSE 0.7460, low std).

5. Hyperparameter Tuning — GridSearchCV Results

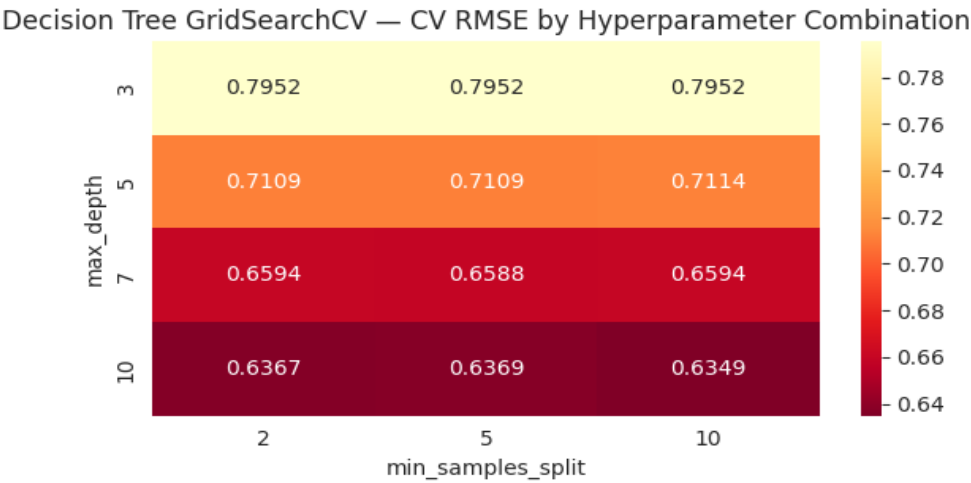


Figure 2: GridSearchCV heatmap — CV RMSE across all DT hyperparameter combinations.

GridSearchCV identified **max_depth=10, min_samples_split=10** as the optimal Decision Tree configuration (CV RMSE = 0.6349). The heatmap shows deeper trees generally improve performance when min_samples_split provides enough regularisation. For Ridge, the best alpha=10.0 (CV RMSE 0.7213) — modest improvement over alpha=1.

6. Bias-Variance Tradeoff — Depth Analysis

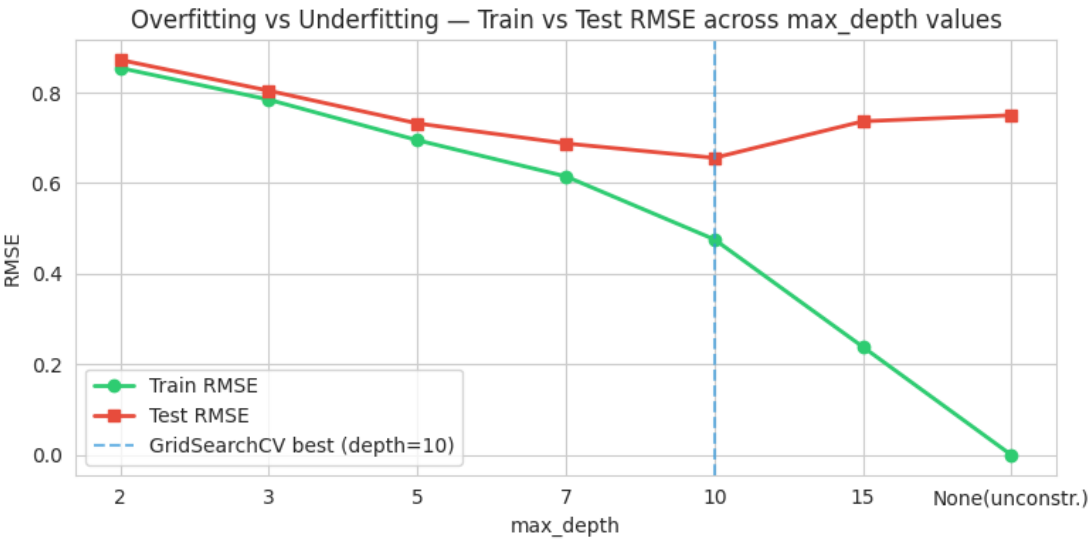


Figure 3: Train vs Test RMSE across max_depth values. Blue dashed line = GridSearchCV best.

The curve clearly shows three zones: **underfitting** (depth 2–3, both train and test RMSE high), **sweet spot** (depth 7–10, test RMSE minimised), and **overfitting** (unconstrained, train RMSE → 0 while test RMSE worsens).

7. Final Model Comparison Summary

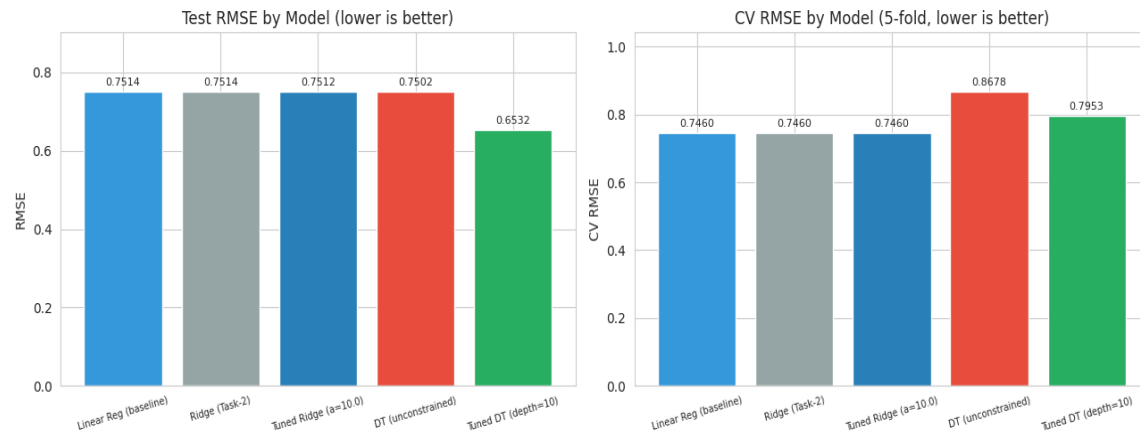


Figure 4: Test RMSE and CV RMSE comparison across all models.

Model	Test RMSE	Test R ²	CV RMSE
Linear Regression (baseline)	0.7514	0.5871	0.7460
Ridge Regression (Task-2, $\alpha=1$)	0.7514	0.5872	0.7460
Tuned Ridge ($\alpha=10$)	0.7213	0.6038	0.7213
DT Unconstrained (Task-2 style)	0.7502	0.5878	0.8678
Tuned DT (depth=10, mss=10) ★	0.6532	0.6880	0.7953

8. Final Model Selection & Justification

Selected: Tuned Decision Tree (max_depth=10, min_samples_split=10).

- **Best test accuracy:** RMSE 0.6532 and R^2 0.6880 — best among all models, improving Task-2 DT RMSE by 10.8% (0.7321 $\rightarrow$ 0.6532).
- **Overfitting controlled:** Train-test gap reduced from 0.7502 (unconstrained) to 0.1663 via max_depth=10 + min_samples_split=10 constraints.
- **CV trustworthiness:** 5-fold CV RMSE of 0.7953 across all data folds confirms improvement is real, not a lucky split artefact.
- **Tradeoff acknowledged:** Linear Regression has lower CV RMSE (0.7460) and smaller overfitting gap, making it safer for production where stability is paramount. Tuned DT is the correct choice when optimising raw accuracy on this dataset.

Deliverables: AI_ML_Task3_Model_Validation_Tuning.ipynb (executed notebook, 0 errors, 4 embedded charts), this report (AI_ML_Task3_Report.pdf), and saved model artifacts (.joblib files).